

Josue Tapia Hernández

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EDUCATION

Tecnológico de Monterrey <i>BS in Mathematics and Data Science</i> – GPA: 97/100 — Academic Scholarship (Top 5%). Coursework: Data Structures & Algorithms, Statistical Inference, Optimization, Deep Learning.	Guadalajara, Mexico <i>Expected June 2027</i>
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EXPERIENCE

AI Engineer Intern <i>True Data Science</i> – Integrating agentic AI into the company’s multi-tenant ticketing service (Spring Boot / PostgreSQL), so that LLM agents can open, comment on, and triage tickets as first-class actors alongside human users. – Automated daily performance reporting for sales managers , replacing a manual process, by building a serverless Python pipeline on AWS (scheduled Lambda, S3 for data, SES for delivery) that renders HTML templates to PDF reports. – Connected internal services end to end (reporting, ticketing, notifications) by building resilient REST integrations with retry/backoff and idempotent writes on DynamoDB.	Aug 2026 – Present Guadalajara, Mexico
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Machine Learning Engineer <i>JATA — AgTech Startup [Website]</i> – Took an AgTech product from hackathon win to 2 paying commercial greenhouse clients across 5 growing seasons , by building and shipping a Dockerized FastAPI backend (7+ routes: auth, ingest, sensors, predictions, admin) on Railway with JWT auth and Postgres row-level security via Supabase. – Let operators track sensor health and forecast-vs-actual yield without engineering support , streaming 16 sensor features at 30s refresh, via a sensor-ingestion service feeding a React dashboard. – Delivered 5-week yield forecasts at $R^2 = 0.70$, MAPE = 10.7% on 365 weekly production records , by training a CNN-RNN (3 residual conv blocks + 3-layer LSTM) served behind the API and covered by pytest integration tests on the ingest, auth, and prediction endpoints.	Sep 2025 – Present Guadalajara, Mexico
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Undergraduate Machine Learning Researcher <i>Tecnológico de Monterrey — Advisor: Prof. Andrés Ramírez Morales</i> – Matched the state-of-the-art model for particle-jet classification (test AUC 0.9811 vs. 0.9813) with $5.7\times$ fewer parameters (189K) , by building a rotation-equivariant transformer from scratch in PyTorch (custom linear, attention, normalization, and MLP layers) with a 60+ test suite that checks the symmetry guarantees numerically. – First author on “ <i>SU(2) Equivariance for Top Jet Classification</i> ” (in preparation, targeting Phys. Rev. D), reaching AUC 0.9850 / 93.8% accuracy on the full 1.2M-sample public benchmark, by running an 8-variant ablation study and building the data pipelines that derive physics features from raw detector measurements.	Dec 2025 – Present Guadalajara, Mexico
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PROJECTS

FLI Holdings — Multi-Agent Financial Copilot [GitHub] Python, Claude API, Pydantic, FastAPI, React/TS, Vite 2026 – Turned 4 months of raw restaurant-group financials (5 branches, 2.7K spreadsheet rows, 44.7% invalid) into executive memos with prioritized action items , by leading a three-agent pipeline with typed Pydantic contracts between agents and a synthesis agent that merges quantitative and qualitative findings into a PDF memo. – Surfaced per-branch KPIs and alerts automatically , by building a quantitative agent that validates the monthly spreadsheets, computes metrics by branch, product, and category, and flags branches below margin or revenue thresholds. – Extracted strategic signals (strengths, risks, growth drivers, sentiment) from unstructured client documents , by building a qualitative agent that ingests PDFs and scanned images via OCR, chunks and embeds them into a vector store (ChromaDB, multilingual sentence embeddings), and runs schema-constrained Claude extraction into typed fields. – Gave analysts reproducible what-if answers (5 payroll/pricing/closure scenarios + break-even ticket solver) with every number LLM-free , verified by 42 tests (pytest + Vitest), by building a FastAPI job API with a thread-safe job store and a React/TypeScript dashboard with KPI overview, “What-If Lab”, and waterfall charts.

Cash-Flow Optimization for Coppel [Report] Python, PuLP (MILP), pandas, Vercel 2025 – Saved \$32.9K MXN/day vs. the unoptimized baseline for moving cash between 21 retail branches , by formulating a mixed-integer linear program (1.3K variables : cash sent per route, armored-truck trips, electronic deposits) that minimizes transport, transfer, insurance, and collection costs while keeping each branch within its allowed cash range, solved on 7 days of real sales data. – Confirmed the plan holds under changing conditions via sensitivity analysis: a ten percent rise in transport tariffs raised total cost by only eight percent, and halving truck capacity added twelve percent more trips while staying feasible; shipped the daily routes and KPIs as a live web dashboard with a map of optimal transfers.

AWARDS & HONORS

1st Place, UST Hackathon — winning entry became JATA, now a revenue-generating AgTech company	2025
ICPC National Regional Contest — Honourable Mention (2 $\times$) — Top 17% nationally	2024, 2025
National Informatics Olympiad — Bronze (Top 15%) — Intl. Olympiad in Informatics (IOI) training	Nov 2022
National Physics Olympiad — Bronze (Top 20%) — Intl. Physics Olympiad (IPhO) training	Nov 2022

SKILLS

Languages: Python, Java, TypeScript, C++, SQL, R, Bash. Spoken: Spanish (Native), English (C1). Backend & Cloud: FastAPI, Spring Boot, REST, JWT/OAuth2, React, PostgreSQL (RLS), Redis, AWS (Lambda, DynamoDB, S3, SES, EventBridge, CDK), Docker, GitHub Actions, Vercel, Railway, pytest, JUnit, Vitest. AI & ML: PyTorch, TensorFlow, scikit-learn, PEFT/LoRA, LangGraph, DSPy, Claude & Gemini APIs, XGBoost, PuLP, Hydra, giotto-tda, geometric algebra.
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